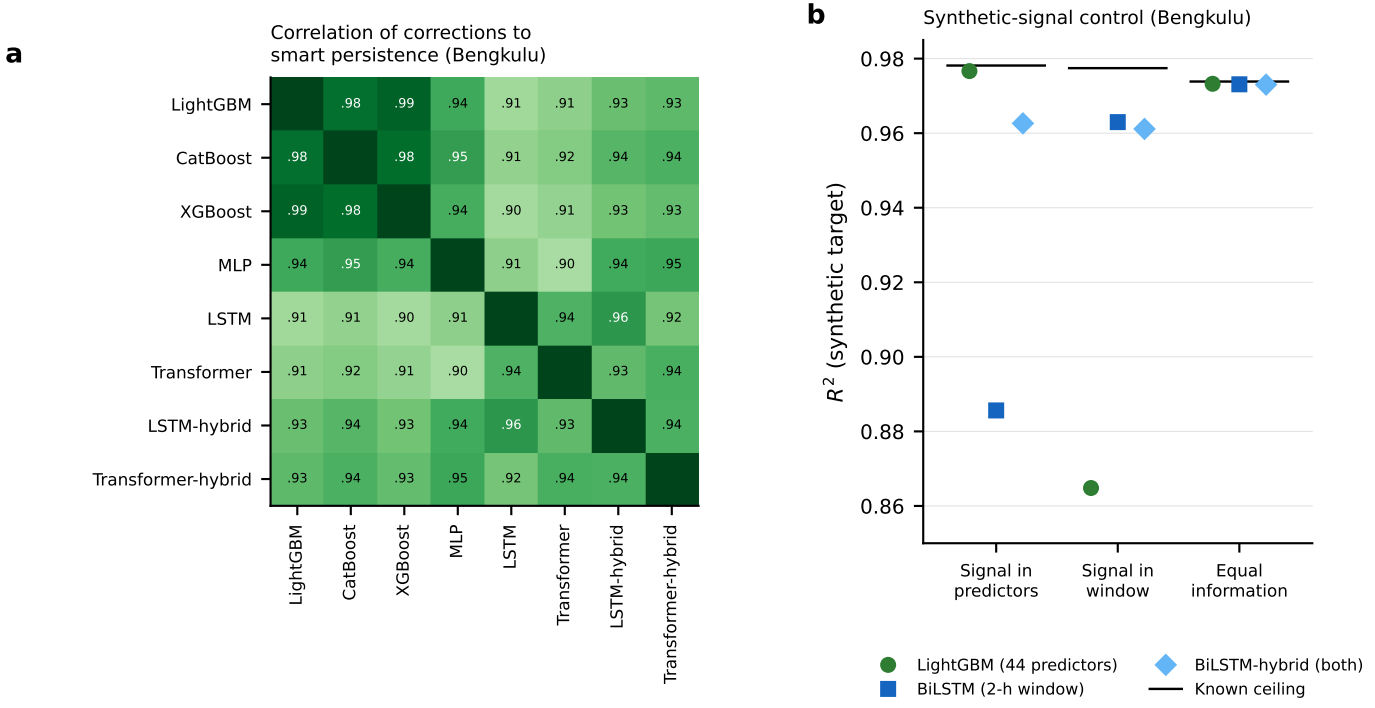
a
Correlation of corrections to smart persistence (Bengkulu)
LightGBM .98 .99 .94 .91 .91 .93 .93
CatBoost .98 .98 .95 .91 .92 .94 .94
XGBoost .99 .98 .94 .90 .91 .93 .93
MLP .94 .95 .94 .91 .90 .94 .95
LSTM .91 .91 .90 .91 .94 .96 .92
Transformer .91 .92 .91 .90 .94 .93 .94
LSTM-hybrid .93 .94 .93 .94 .96 .93 .94
Transformer-hybrid .93 .94 .93 .95 .92 .94 .94
LightGBM
CatBoost
XGBoost
MLP
LSTM
Transformer
LSTM-hybrid
Transformer-hybrid
b
Synthetic-signal control (Bengkulu)
R^2 (synthetic target)
0.98
0.96
0.94
0.92
0.90
0.88
0.86
Signal in predictors
Signal in window
Equal information
LightGBM (44 predictors)
BiLSTM (2-h window)
BiLSTM-hybrid (both)
Known ceiling